

The Discovery of GSK3227634: A Potent Inhaled Inhibitor of the KEAP1/NRF2 Protein–Protein Interaction for the Treatment of Chronic Obstructive Pulmonary Disease

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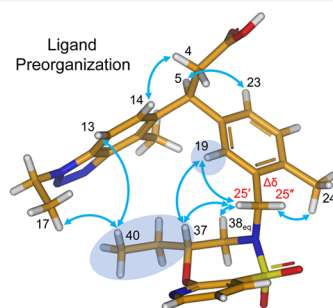
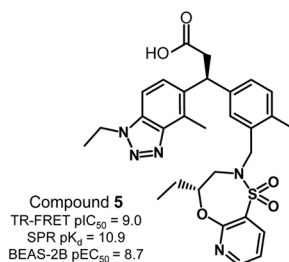
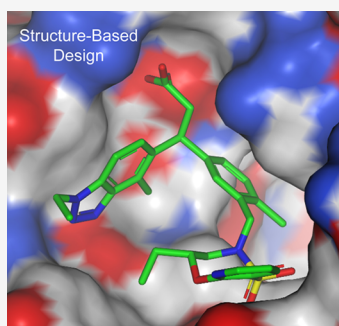
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ABSTRACT: KEAP1 is the key regulator of the NRF2-mediated cytoprotective response and a target for pathologies involving oxidative stress. Compounds that covalently modify KEAP1 to activate NRF2 are clinically validated; however, their chemical reactivity may drive increased off-target activity. A more selective approach involves inhibition of the KEAP1–NRF2 protein–protein interaction as exemplified by our previous bis-aryl lead KI-696 (4). We now describe the lead optimization of the bis-aryl series that focused on increasing the population of the bioactive conformation of the free ligand to drive potency while optimizing overall physicochemical properties. This resulted in the discovery of GSK3227634 (5), an ultrahigh-affinity noncovalent inhibitor of KEAP1–NRF2 (surface plasmon resonance pK_d = 10.9) and the first NRF2 activator to demonstrate target engagement and efficacy in preclinical models of oxidative stress via direct delivery to the lung. Compound 5 therefore represents a novel potential agent to treat lung diseases involving oxidative stress, such as chronic obstructive pulmonary disease.

INTRODUCTION

Chronic obstructive pulmonary disease (COPD) is a heterogeneous age-related degenerative condition that principally manifests as irreversible airflow limitation and progressive loss of lung function with predominant respiratory symptoms of dyspnea and chronic cough. Despite the existence of effective, symptom-modifying therapies such as inhaled corticosteroids and bronchodilators, COPD is the seventh leading cause of poor health worldwide as measured by disability-adjusted life years.¹ Oxidative stress has long been considered one of the major drivers of COPD pathogenesis and disease progression.² Endogenous and exogenous oxidative stressors can lead to an increase in reactive oxygen species (ROS) that ultimately lead to impaired cellular function and structure, eventually resulting in

airway remodeling and lung destruction.³ The lung is equipped with antioxidant response systems that attenuate excessive ROS, and key among these is nuclear factor erythroid 2-related factor 2 (NRF2), a transcription factor that regulates the production of many detoxifying, antioxidant, and cytoprotective proteins. Often described as a master regulator of oxidative stress, the

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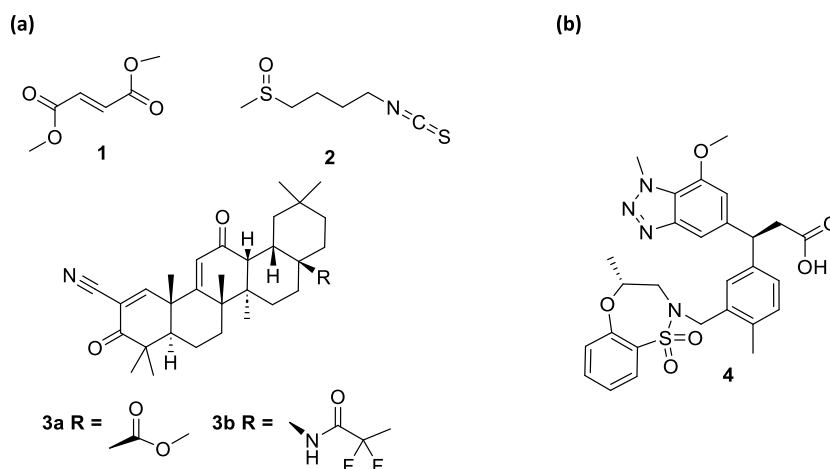


Figure 1. Examples of NRF2 activators that (a) covalently or (b) noncovalently interact with KEAP1.

Table 1. Structure–Activity Relationships for Bis-Aryl Series NRF2 Activators^a

Cpd	R ₁	R ₂	R ₃	R ₄	C5	KEAP1 FP pIC ₅₀ (n)	NRF2-KEAP1 LP TR-FRET pIC ₅₀ (n)	BEAS2B pEC ₅₀ (n)
4	OMe	H	Me	(R)-Me	S	7.7 (16)	8.9 (10)	8.0 (23)
6	OMe	H	Me	(R)-Me	R	7.6 (18)	8.3 (10)	7.1 (20)
7	OMe	H	Et	(R)-Et	S	7.6 (2)	8.5 (2)	8.8 (2)
8	OMe	H	Et	(R)-Et	R	7.6 (2)	8.5 (2)	8.2 (2)
9	H	Me	Me	(R)-Me	S	7.7 (40)	9.0 (34)	8.2(44)
10	H	Me	Me	(R)-Me	R	7.9 (8)	8.7 (4)	8.0 (6)
11	H	Me	Me	(R)-Et	S	7.7 (6)	9.0 (4)	9.0 (8)
12	H	Me	Me	(R)-Et	R	7.6 (8)	8.9 (4)	8.6 (12)
13	H	Me	Et	(R)-Et	S	7.7 (4)	9.0 (4)	9.1 (4)
14	H	Me	Et	(R)-Et	R	7.7 (4)	8.9 (4)	8.6 (4)

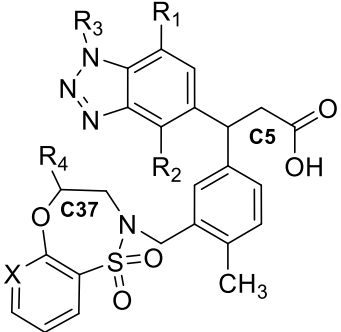
^a(n) = number of determinations.

NRF2 pathway is responsible for maintaining cellular redox homeostasis during events resulting in injury, inflammation, and infection.⁴ NRF2 is mainly regulated by its interaction with the Kelch-like ECH-associated protein 1 (KEAP1)-Cullin3 (Cul3)-based E3-ubiquitin ligase complex. Under basal conditions, NRF2 binds to the Kelch domain of the KEAP1 protein, facilitating the ubiquitination and subsequent degradation of cytoplasmic NRF2 by the E3-ubiquitin ligase Cul3, which keeps NRF2 concentrations in the cytoplasm at a low level.^{4b} However, in the presence of oxidative stress or electrophiles, modification of cysteine residues in the Bric-a-Brac (BTB) domain and the Intervening Region (IVR) of KEAP1 inactivates its function as a substrate adaptor, resulting in the concomitant rise in nuclear levels of NRF2. Subsequently, NRF2-regulated gene transcription is upregulated with the increased expression of cytoprotective enzymes and proteins (e.g., NAD(P)H quinone dehydrogenase 1 (NQO1), heme oxygenase (HO-1), glutathione (GSH), sulfiredoxin-1 (SRXN1), and thioredoxin

reductase 1 (TXNRD1)) to broadly protect mitochondrial and cell health.⁵

Small molecules that covalently modify KEAP1 cysteine residues integral for the oxidative stress response and activate the NRF2 pathway have shown encouraging clinical results⁶ (e.g., dimethyl fumarate 1 approved for multiple sclerosis), sulforaphane 2, bardoxolone methyl 3a, and omaveloxolone 3b (approved for Friedreich's ataxia) (see Figure 1a).

However, inherent to their modes of action, these compounds feature reactive functionalities that have the potential for increased off-target activity, which may affect their therapeutic index. An alternate, more selective approach toward NRF2 activation involves noncovalent, reversible inhibition of the protein–protein interaction (PPI) between NRF2 and the Kelch domain of KEAP1 based on either peptides or small molecules.⁷ NRF2 binds two Kelch domains of dimeric KEAP1 through high-affinity (“ETGE”) and low-affinity (“DLG”) motifs that interact with a pocket at the center of a β -propeller

Table 2. Structure–Activity Relationships for Bis-Aryl Series NRF2 Activators^a


Cpd	R ₁	R ₂	R ₃	R ₄	X	C5	KEAP1 FP pIC ₅₀ (n)	NRF2-KEAP1 LP TR-FRET pIC ₅₀ (n)	BEAS2B pEC ₅₀ (n)
5	H	Me	Et	(R)-Et	N	S	7.8 (14)	9.0 (12)	8.7 (18)
7	OMe	H	Et	(R)-Et	CH	S	7.6 (2)	8.5(2)	8.8 (2)
8	OMe	H	Et	(R)-Et	CH	R	7.6 (2)	8.5 (2)	8.2 (2)
15	OMe	H	Et	(R)-Et	N	S	7.7 (2)	8.7 (2)	8.6 (2)
16	OMe	H	Et	(R)-Et	N	R	7.8 (2)	8.9 (2)	8.1 (2)
17	OMe	Me	Et	(R)-Et	N	S	7.6 (6)	8.9 (6)	9.4 (6)
18	OMe	Me	Et	(R)-Et	N	R	7.6 (4)	9.1 (4)	8.7 (4)
19	H	Me	Et	(R)-Me	N	S	7.7 (2)	9.1 (2)	8.1 (2)
20	H	Me	Et	Me, Me	N	S	7.5 (6)	9.0 (6)	7.5 (6)
21	H	Me	Me	(R)-Et	N	S	7.6 (4)	9.1 (4)	8.4 (4)
22	H	Me	Et	(R)-Et	N	R	7.8 (18)	9.0 (14)	8.1 (23)
23	H	Me	Et	(S)-Et	N	S	7.8(10)	8.7 (8)	7.5 (10)
24	H	Me	Et	(S)-Et	N	R	7.7 (64)	8.2 (74)	7.1 (108)

^a(n) = number of determinations.

fold. Loss of one or both interactions through the action of PPI inhibitors has been shown to prevent the productive engagement of NRF2 with KEAP1, with subsequent activation of its antioxidant transcriptional pathway.⁸ We have reported previously the application of fragment-based drug discovery to the Kelch domain of KEAP1, which included the exploration and optimization of protein–ligand interactions in three energetic “hot spots” identified by fragment screening and leading to the discovery of the bis-aryl series of PPI inhibitors exemplified by the bis-aryl tool compound KI-696 (**4**, Figure 1b).⁹ Subsequently, we reported a detailed account of the medicinal chemistry campaign leading to **4**, which displayed tight, selective, nanomolar binding to the Kelch domain and the ability to activate the NRF2 pathway both in vitro and in vivo.¹⁰ This current report now describes the subsequent lead optimization of the bis-aryl series resulting in the discovery of GSK3227634 (**5**), an ultrahigh affinity, noncovalent, reversible inhibitor of the binding of NRF2 to KEAP1 and activator of the NRF2 pathway. One of the key aspects of the lead optimization effort focused on increasing the bioactive conformation of the free ligand to drive potency, and we present here solution-phase and crystallographic data that address this. In addition to displaying in vivo activation of the NRF2 pathway after intravenous (IV) administration, improvements in affinity and cellular efficacy have resulted in **5** being the first NRF2 activator to demonstrate robust target engagement and efficacy following direct administration to the lung. Thus, compound **5** has the potential to treat acute diseases amenable to continuous IV administration. More importantly, compound **5** displays an overall profile supporting inhaled delivery useful for the treatment of chronic lung diseases, like COPD.

RESULTS

Structure–Activity Studies

The key objectives for the lead optimization campaign were to further increase affinity of bis-aryl series analogues for the Kelch domain of KEAP1, improve their cellular efficacy, and to balance their physicochemical properties (permeability, solubility, and crystallinity) in order to achieve a sufficiently low dose to support a formulation suitable for chronic delivery by inhalation. A fluorescent polarization (FP) assay was used to drive the early progression of the inhibitors; however, we found that more optimized compounds in this series reached the ceiling of this assay, with all analogues displaying affinities in the 16–25 nM (pIC₅₀ = 7.6–7.8) range (Tables 1 and 2). We therefore developed a more sensitive low-protein time-resolved fluorescence resonance energy transfer (LP TR-FRET) assay that was used to compare the affinity of the analogues for KEAP1 by measuring the inhibition of the interaction between full-length KEAP1 and full-length NRF2. The inhibition data for **4** in this LP TR-FRET assay (pIC₅₀ = 8.9) are in good agreement with the previously determined K_d for **4** (1.3 nM) measured by isothermal titration calorimetry.⁹ A human cell-based assay was utilized to determine engagement of the NRF2 pathway. In human bronchial epithelial cells (BEAS-2B), the enzymatic activity of NQO1, a known downstream effector of NRF2, increased stepwise in response to increasing amounts of NRF2 activator. The apparent potency value (pEC₅₀) was measured after 48 h of incubation, which was determined as the temporal optima for this assay.

Lead optimization of the bis-aryl series was driven by knowledge of the previously determined crystal structure of **4** bound to the KEAP1 Kelch domain, in conjunction with observations presented in our previous publications.^{9,10} The

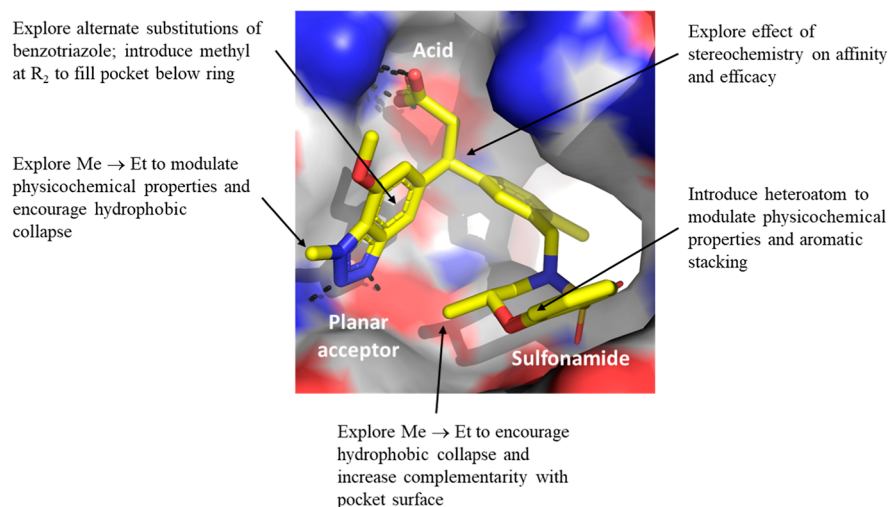


Figure 2. Lead optimization design strategy for bis-aryl compounds in the context of the previously determined structure of **4** bound to the Kelch domain of KEAP1 (PDB 5FNU).

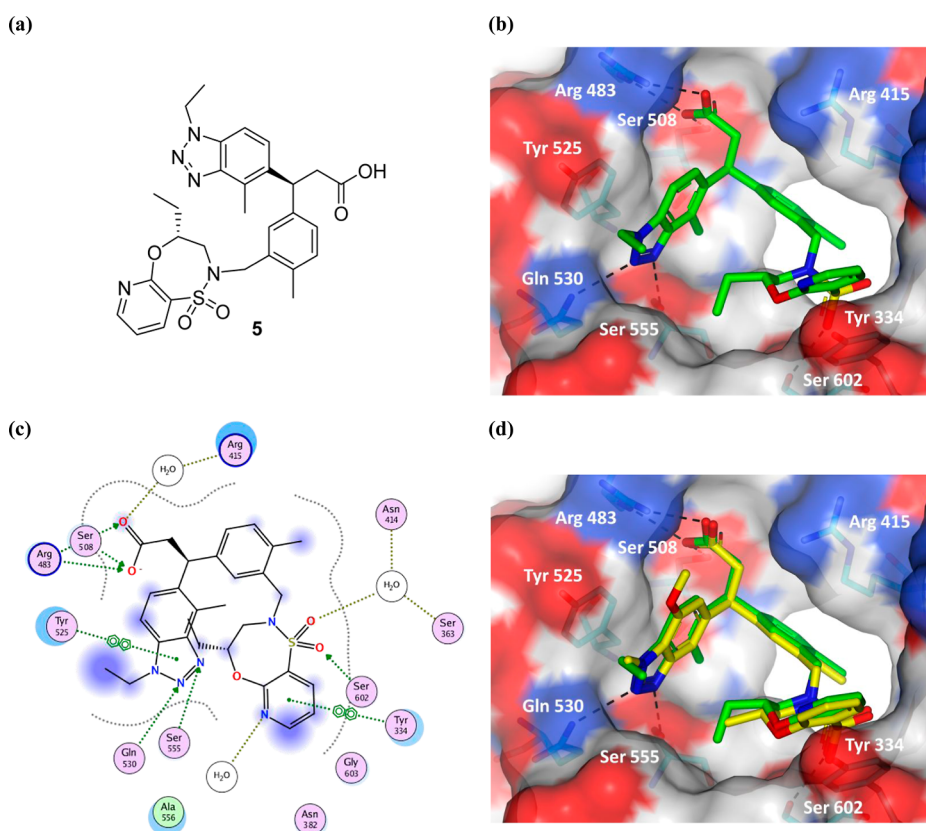


Figure 3. X-ray structures of bis-aryl series compounds bound to Kelch domain of KEAP1. (a) Compound **5**; (b) compound **5** bound to KEAP1. Hydrogen bonds and electrostatic interactions are denoted as dashed lines (PDB 30HE); (c) key interactions of compound **5** with KEAP1; and (d) overlay of compounds **4** (yellow, PDB 5FNU) and **5** (green) bound to KEAP1.

optimization strategy of **4** focused on three main areas of the molecule, namely, the benzotriazole, the oxathiazepine ring, and the aryl group fused to the oxathiazepine. Key elements of the design are summarized in Figure 2. They included exploration of small lipophilic substituents at R_2 (see Tables 1 and 2) on the benzotriazole, in order to fill space and increase surface complementarity; introduction of heteroatoms into the fused aryl ring to modulate physicochemical properties and π -stacking; and introduction of ethyl substituents at R_3 and R_4 .

These homologations of methyl substituents to ethyl were in part designed to further encourage the propensity for the “hydrophobic collapse” reported previously for **4**, increasing the proportion of the bioactive conformation for the free ligand and potentially providing a route to the very high affinity and efficacy desired for inhaled dosing (see Conformational Studies of **5** below).¹⁰ In addition, the homologation to ethyl at R_4 provides an opportunity to further increase surface complementarity between the ligand and the NRF2-binding site at this point.

Table 3. Profiles of Selected Bis-Aryl Series NRF2 Activators^a

Cpd	NRF2-KEAP1 LP TR-FRET pIC ₅₀ (n)	BEAS2B pEC ₅₀ (n)	chrom log D pH7.4	AMPA pH 7.4 (nm/s)	MDCK perm (nm/s) [efflux ratio]	PPB rat	in vivo PK rat	
							Cl (mL/min/kg)	V _{ss} (L/kg)
4	8.9 (10)	8.0 (23)	3.08	32	1.5 [>47]	99.5	38	0.9
5	9.0 (12)	8.7 (18)	2.55	35	2.3 [>49]	98.9	50	0.5
11	9.0 (4)	9.0 (8)	3.26	72	4.4 [111]	99	46	0.5
12	8.9 (4)	8.6 (12)	3.24	103	2.4 [>120]	99.5	56	0.6
22	9.0 (14)	8.1 (23)	2.49	46	1.2 [>52]	98.6	47	0.4

^aAMPA = artificial membrane permeability assay; PPB = plasma protein binding; V_{ss} = volume of distribution at steady state; and (n) = number of determinations.

Throughout lead optimization, we sought to maintain the electrostatic and hydrogen bonding motifs observed for **4** in the “Acid”, “Planar acceptor”, and “Sulfonamide” hot spots shown in Figure 2, as these were known to be important for activity from our earlier studies.^{9,10}

We now describe the structure–activity relationships (SARs) associated with our optimization of the bis-aryl series in further detail. As described in our previous publications, the S-isomer at C5 of the bis-aryl template is usually preferred (e.g., compare compounds **4/6**, **7/8**, **9/10**, **11/12**, and **13/14**), with the R-isomer displaying both lower affinity and cellular efficacy (Table 1).^{9,10} Replacing methyl with ethyl at R₃ and R₄ in **4** provided **7**, which retained affinity for KEAP1 while increasing cellular efficacy. Removal of the OMe substituent at R₁, while incorporating a methyl group at R₂ to fill space below the benzotriazole ring, afforded **9**, **11**, and **13** with limited effects on affinity. However, analogues with R₄ = Et, **11** and **13**, display increased cellular efficacy when compared to that of R₄ = Me, **9**.

To modify the overall physicochemical properties of the series, we introduced a nitrogen into the cyclic aryl sulfonamide ring, which stacks against Tyr 334 at the edge of the NRF2-binding pocket. As described in our previous publication, substitution of the aromatic system at this position to modulate electronics exhibited relatively flat SAR, providing an opportunity to explore overall physicochemical properties such as lipophilicity while maintaining potency.¹⁰ In agreement with this, the change to give compound **15** afforded only limited effect on affinity and efficacy when compared to **7** (Table 2). Replacing the methoxy at R₁ with a methyl group at R₂ in this context again had limited effect on affinity or cellular efficacy, but methyl was preferred going forward, in part, to minimize overall molecular weight. As in the previous analogues, incorporation of ethyl substituents at both R₃ and R₄ provided the best cellular efficacy, as exemplified by **5** (Figure 3a), possibly due to a combination of ligand preorganization and modulation of lipophilicity (see below). As observed previously, substitution with the correct stereochemistry at C37 is important for both affinity and cellular efficacy, with the R-isomer preferred over either the S-isomer, **23** and **24**, or a gem-dimethyl substituent, **20**.¹⁰

In order to confirm the binding mode of **5**, we determined its crystal structure bound to the Kelch domain of KEAP1 (Figure 3b), and a summary of key protein–ligand interactions formed in the complex is shown in Figure 3c. The ligand binding site for **5** is located at the narrow end of the beta propeller structure of KEAP1 and lined by residues Tyr 334, Ser 363, Gly 364, Asn 382, Arg 415, Ile 461, Gly 462, Phe 478, Arg 483, Ser 508, Gly 509, Tyr 525, Gln 530, Ser 555, Ala 556, Tyr 572, Phe 577, Ser 602, Gly 603, and a number of well-ordered water molecules. The carboxyl group of **5** forms an electrostatic interaction with

Arg 483 and accepts a hydrogen bond from Ser 508, as observed in the structure of **4** with KEAP1. The sulfonyl group forms hydrogen bonds with Ser 602 and a water molecule held in place by interactions with the side chains of Ser 363 and Asn 414. The benzotriazole accepts hydrogen bonds from the side chains of Ser 555 and Gln 530 as well as forming a π -stacking interaction with Tyr 525.

The pyridine ring also forms a π -stacking interaction with Tyr 334 at the lipophilic edge of the pocket. In general, there was little perturbation to the overall binding mode of **4** during optimization to **5**, with key patterns of hydrogen bonding, electrostatic interactions, and aromatic stacking maintained throughout this process (Figure 3d). Thus, the key points of the KEAP1 pharmacophore identified from the fragment screen, namely, “Acid”, “Sulfonamide”, and “Planar acceptor” are maintained within this more optimized molecule.

Two sets of diastereomeric isomers at C5, **11/12** and **5/22**, along with compound **4** were selected for further in vitro and in vivo evaluation. Their profiles are summarized in Table 3.

All of these analogues displayed high affinity for KEAP1 as measured in the LP TR-FRET assay. Compounds **5** and **11** with the S-configuration at C5 also displayed greater efficacy in the cellular assay when compared to **4** or their corresponding R-isomers, **12** and **22**. The introduction of a nitrogen in the cyclic aryl sulfonamide ring decreased lipophilicity as determined by chromatographic log D (chrom log D). These compounds generally display low permeability with high efflux ratios in Madin–Darby canine kidney (MDCK) cells, low free fractions, and high in vivo clearance (Cl) in the rat after IV administration. The limited permeability and high systemic clearance profiles of these analogues are advantageous for local delivery of these compounds to the lung via inhalation. Targeting the lung directly provides the potential to attenuate any systemic toxicity issues. Also, the oral bioavailability of **5** was found to be relatively low and variable in an intravenous/oral pharmacokinetic crossover study. In addition, in preliminary rat IT screening studies, **5** was the only lead analogue that demonstrated significant activity at the single screening dose utilized. Thus, taking the overall in vitro and in vivo profiles of **5** into consideration with the major objective of progressing a compound that supports the inhaled route of administration, **5** was selected as the molecule for further profiling.

Surface Plasmon Resonance Characterization of Compound 5

As the affinity of **5** for the Kelch domain of KEAP1 was anticipated to have reached the ceiling even in the LP TR-FRET assay, its binding was further evaluated using surface plasmon resonance (SPR). Due to the high affinity and slow off-rate, the measurement was configured to monitor dissociation over a 4 h period to confirm reversibility and measure an approximate

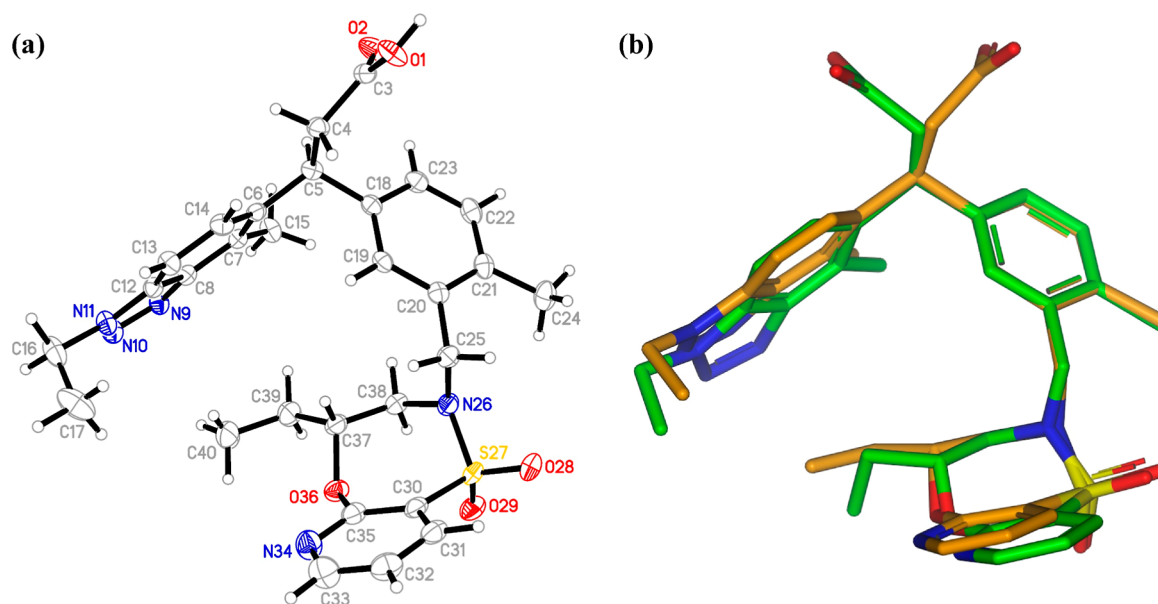


Figure 4. Conformation of **5** in the small molecule and protein–ligand crystal structures. (a) Conformation for **5** in the small molecule crystal structure. Anisotropic atomic displacement ellipsoids for the nonhydrogen atoms are shown at the 50% probability level. Hydrogen atoms are displayed with an arbitrarily small radius. (b) Overlay of KEAP1-bound **5** (green) and small molecule structure (orange). Superposition was based on atoms in the tolyl system. Hydrogens have been omitted for clarity.

dissociation rate constant and residence time. A set of three determinations with **5** yielded a pK_d of 10.9 ± 0.6 with a $t_{1/2}$ of approximately 4 h confirming very tight binding to the Kelch domain of KEAP1. For comparison, the affinity of a NRF2 peptide (AFFAQLQLDEETGEFL) yielded a pK_d of 7.2 ± 0.04 with a $t_{1/2} = 25$ s. While the data show that **5** is reversible yet very slow to dissociate, the measured $t_{1/2}$ is not precisely defined by the data given the lack of significant curvature of the dissociation phase of the kinetics (see [Supporting Information](#), Figure S2).

Conformational Studies of **5**

Previously, we presented evidence from NMR studies indicating a major solution phase conformation for the tool compound **4**, which is similar to that bound to KEAP1.¹⁰ In particular, we demonstrated that the specific addition of a methyl group to the oxathiazepine was associated with an increased population of the bioactive conformation, most likely due to a combination of local conformational restriction and “hydrophobic collapse” between the benzotriazole heterocycle, the central phenyl and the seven-membered ring. We hypothesized that this preorganization might be partially responsible for its increased affinity over the des-methyl analogue, as well as providing an explanation for its excellent efficacy in the BEAS2B cell-based assay.¹⁰

The homologation of methyl to ethyl in analogues including **5** was in part designed to further encourage the propensity for the hydrophobic collapse observed for **4** and increase the proportion of the bioactive conformation of the free ligand. To gain insight into any potential conformation preference for **5**, we initially determined its small molecule crystal structure and compared it with that bound to KEAP1. The conformation for **5** from the small molecule determination is presented in [Figure 4a](#) and the structure is shown overlaid with its KEAP1-bound pose in [Figure 4b](#). This analysis shows that **5** exhibits a very similar conformation in the small molecule crystal structure to that bound to KEAP1, potentially indicating a degree of preorganization for the free ligand. The relative disposition of the benzotriazole, central phenyl, and oxathiazepine ring systems

is essentially identical in both the small molecule and protein-bound crystal structures, with **5** also adopting the “collapsed” bioactive conformation within the context of the small molecule structure. The C20–C25–N26 methylene linker overlays well for both KEAP1-bound and small molecule structures, and the seven-membered ring adopts an identical pseudochair conformation with the ethyl substituent in the equatorial position. The main point of difference is an alternative conformation for the carboxylic acid group, which adopts a distinct rotamer for the small molecule structure compared to that bound to the protein, and minor torsional changes for the ethyl and benzotriazole moieties. These small differences, which are not expected to significantly differ in energy, may be indicative of some flexibility in these parts of the molecule, albeit within the context of overall preorganization to the collapsed conformation.

In order to understand whether the small molecule and protein–ligand crystal structures of **5** also reflect a dominant conformation of **5** in solution, and to allow a direct comparison with our previous solution-phase studies of this chemical series, we complemented the crystallographic data with NMR studies of the free ligand. As described for **4**, an initial observation from the ¹H 1D spectrum of **5** in aqueous solution was that the diastereotopic protons on the phenyl-sulfonamide methylene linker (25' and 25'', [Figure 5a,b](#)) gave widely separated chemical shifts, with a $\Delta\delta$ of 0.65 ppm (cf 0.59 ppm for **4**¹⁰), consistent with restricted rotation around the C20–C25 and/or C25–N26 bonds ([Figure 5c](#)). In addition, and again as observed for the analogous protons in **4**, ¹H–¹H ROESY data collected in aqueous solution showed different through-space correlations for 25' and 25'', in agreement with a preferred orientation relative to the phenyl and oxathiazepine rings, which is similar to that observed in the small molecule crystal structure and KEAP1-bound conformation ([Figure 5b](#) and [Supporting Information](#)). Distinct equatorial (2.87, d, ²J = 15.6 Hz) and axial (3.64, dd, ²J = 15.6, ³J_{ax-ax} = 10.3 Hz) signals were also observed for 38-H₂ in the seven-membered ring, consistent with

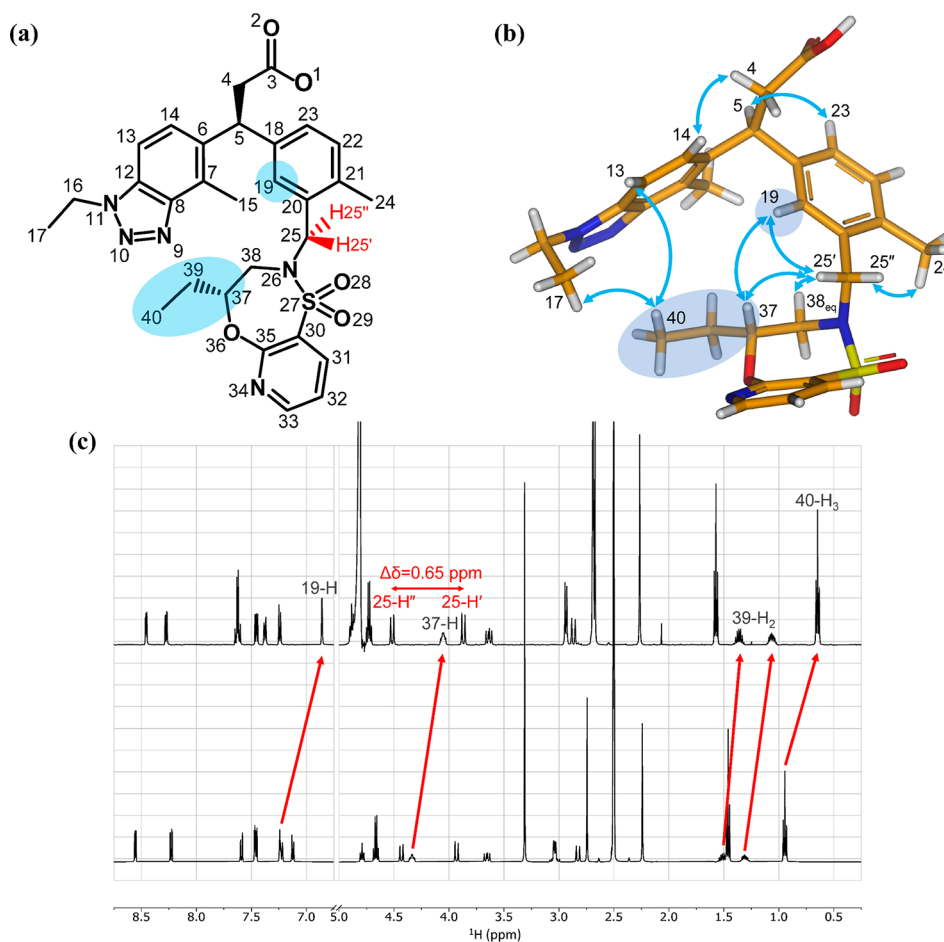


Figure 5. Solution-phase NMR study of free **5**. (a) Chemical structure of **5**. Hydrogens 25' and 25'' on the methylene linker are highlighted in red, and regions showing significant upfield shifts in aqueous solution are highlighted in blue. (b) Conformation of **5** as observed in the small molecule crystal structure. Key ROEs consistent with this conformation are depicted as blue arrows, and regions showing significant upfield shifts consistent with shielding in aqueous solution are highlighted in blue. (c) ¹H NMR spectra for **5** in aqueous solution (top) and DMSO (bottom). Key resonances referred to in text are highlighted, and those undergoing significant (>0.1 ppm) upfield shifts between the two media are highlighted with red arrows.

a single pseudochair conformation with the ethyl substituent pseudoequatorial as observed by crystallography.

Having identified features in the solution data consistent with local preorganization in the methylene linker and seven-membered ring, we next examined the ¹H–¹H ROESY and ¹H 1D data for features consistent with the collapsed conformation observed in the small molecule and protein-bound crystal structures. Two signatures in the NMR spectra of **5** were identified, which suggested a major contribution from this conformation. First, the terminal methyl of the oxathiazepine's pendant ethyl group (40-H₃) shows ¹H–¹H through-space correlations with the N-ethyl benzotriazole substituent (17-H₃), and with the benzotriazole aromatic hydrogen 13-H, in agreement with a similar disposition of these moieties in the free ligand compared to that in the small molecule crystal structure (Figure 5b and Supporting Information). Second, the 40-H₃ methyl protons exhibit a chemical shift (0.65 ppm), which is ~0.2 ppm upfield relative to corresponding resonances for the model compounds propan-1-ol and butan-2-ol.¹¹ This difference in chemical shift is consistent with shielding arising from its position on top of the benzotriazole ring as observed in the small molecule structure.

In order to examine further the propensity for the collapsed versus "extended" conformations, we compared the ¹H spectra for **5** in aqueous solution with that acquired in DMSO (Figure

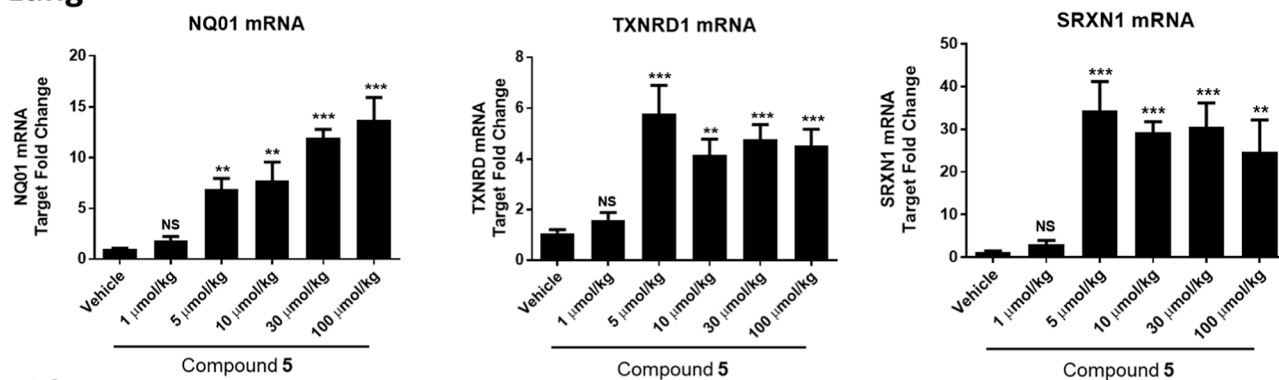
5c). The spectra show a large number of similarities, but a distinct pattern of shielding was observed for a subset of resonances in aqueous solution, namely, those in the oxathiazepine-ethyl CH–CH₂–CH₃ moiety (37-H, 39-H₂, and 40-H₃) and the aromatic hydrogen 19-H from the central phenyl (Figure 5a–c). As these protons would change environment to lie above the benzotriazole in the collapsed (versus extended) state, we propose that this might result from an increased population of this particular conformation in aqueous media. This agrees with our hypothesis that the collapsed solution-phase conformation of **5** is driven by the hydrophobic effect as suggested previously for **4**.¹⁰

In summary, the conformation of **5** observed by crystallography accounts for a large number of ¹H–¹H through-space correlations and shielding effects observed by solution NMR. Taken together, the solution phase study and the small molecule X-ray crystallographic data suggest that free **5** adopts a major conformation, which is very similar to the bioactive conformation bound to KEAP1, and in agreement with studies of **4** and related compounds, which demonstrated the propensity for preorganization in this series.¹⁰

Selectivity Profiling of **5**

The selectivity of **5** was evaluated in a screen of receptor and enzyme assays that have the potential to pose safety issues. Only

(a) Lung



(b) Kidney

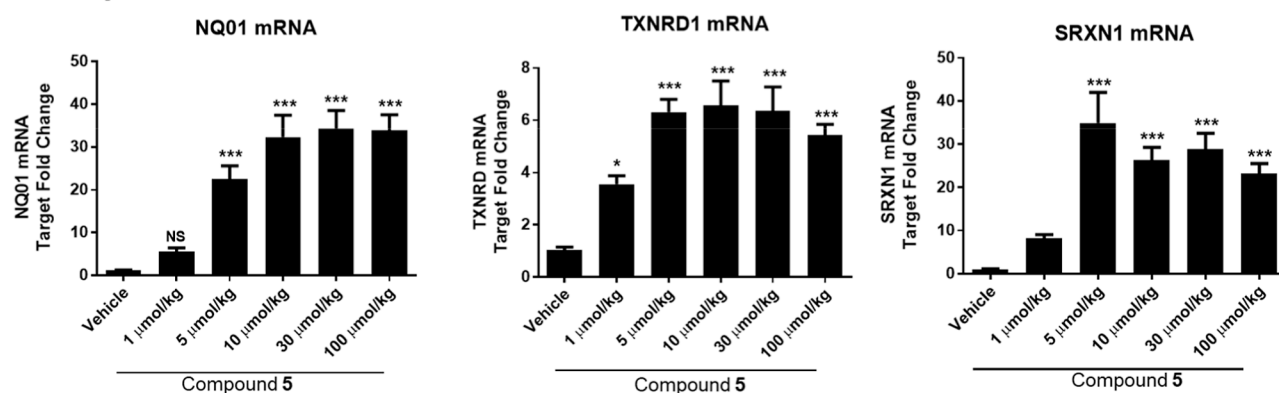


Figure 6. In vivo activation of the NRF2 pathway by compound **5** (IV infusion, 6 h). Compound **5** induces dose-dependent increases in the expression of NRF2-regulated genes in rat lung (a) and kidney (b) tissues in vivo (mean $\pm$ SEM). Statistical analysis: one-way ANOVA with Dunnett's correction; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ versus vehicle.

four of the targets displayed a $\text{pXC}_{50} \geq 5.0$, and, in all these cases, the observed pXC_{50} was ≤ 5.6 .¹² Additional selectivity evaluation was carried out utilizing affinity enrichment chemo-proteomics (AEC) and thermal proteome profiling (TPP). For AEC, **5** and a functionalized analogue of **5** (Supporting Information, compounds **S1** and **S2**, respectively) were immobilized on sepharose beads. These matrices were used to capture endogenous proteins from cell extracts, and competition binding experiments with **5** were performed as described by Eberl et al.¹³ F-box and leucine-rich repeat protein 17 (FBXL17), KEAP1, Kelch-like protein 2 (KLHL2), Kelch-like protein 7 (KLHL7), Kelch-like protein 17 (KLHL17), phosphoglycerate mutase family member 5 (PGAM5), and sequestosome-1 (SQSTM1) were affinity-captured by both matrices and binding was inhibited by excess concentrations of **5** in a dose-dependent manner in replicate experiments. For KEAP1 itself, a pK_d^{app} of 9.4 was determined (Supporting Information, Table S12). PGAM5 and SQSTM1 are well precedented interactors of KEAP1¹⁴ and it is reported that KEAP1 can also form complexes with FBXL17 and members of the KLHL family of proteins, so these proteins are most likely copurified with KEAP1, rather than binding **5** directly.¹⁵ Not all proteins displayed the same apparent affinities for **5** as KEAP1, and this might be explained by **5** modulating the Kelch domain differently depending on the complex composition. The inability to demonstrate direct binding of **5** to KLHL2 by NMR (Supporting Information, Figure S13) supports the hypothesis that **5** may be interacting with heterodimeric complexes of KEAP1 and KLHL proteins.

The selectivity of **5** was also assessed by 2-dimensional TPP based on the cellular thermal shift assay principle on a proteome

wide scale.¹⁶ In this experiment, HepG2 cells were incubated with four different concentrations of **5**, and isothermal dose-response curves were obtained at 12 temperatures. KEAP1 showed full stabilization even at the lowest assay concentration ($\text{pEC}_{50} > 7.7$). Dose-dependent stabilization effects were observed for three additional proteins: fatty acid-binding protein 1 liver (FABP1) ($\text{pEC}_{50} = 6.7$), Yip1 family member 3 (YIPF3) ($\text{pEC}_{50} = 6.1$), and tubulin, beta class I (TUBB) ($\text{pEC}_{50} = 6.0$). In addition, stabilization effects (>2 -fold) were also observed for Aldo-keto reductase family 1, member C3 (AKR1C3), adenine phosphoribosyltransferase (APRT), coproporphyrinogen oxidase, CoA synthase (COASY), and protoporphyrinogen oxidase (PPOX) at the highest compound concentration used in the assay (Supporting Information, Table S13). Low-abundance and substoichiometric complex components of KEAP1 were not detected in the assay. In summary, **5** potently binds to KEAP1 complexes with potencies indicating complex-specific binding. FABP1 was detected as a potential off-target, although it displays a substantially lower potency.

In Vivo Pharmacological Profiling of **5**: Systemic Administration

Initially, due to its high in vivo clearance and very low oral exposure, **5** was evaluated in vivo in rats via IV infusion to demonstrate engagement of the NRF2 pathway. Han Wistar rats were administered **5** by IV infusion over 6 h, and lung and kidney tissues were analyzed for expression of NRF2-regulated genes. In the lung, NQO1 expression increased in a dose-dependent manner from 1 to 100 $\mu\text{mol/kg}$, while expression levels of TXNRD1 and SRXN1 reached a maximum at 5 $\mu\text{mol/kg}$. In the

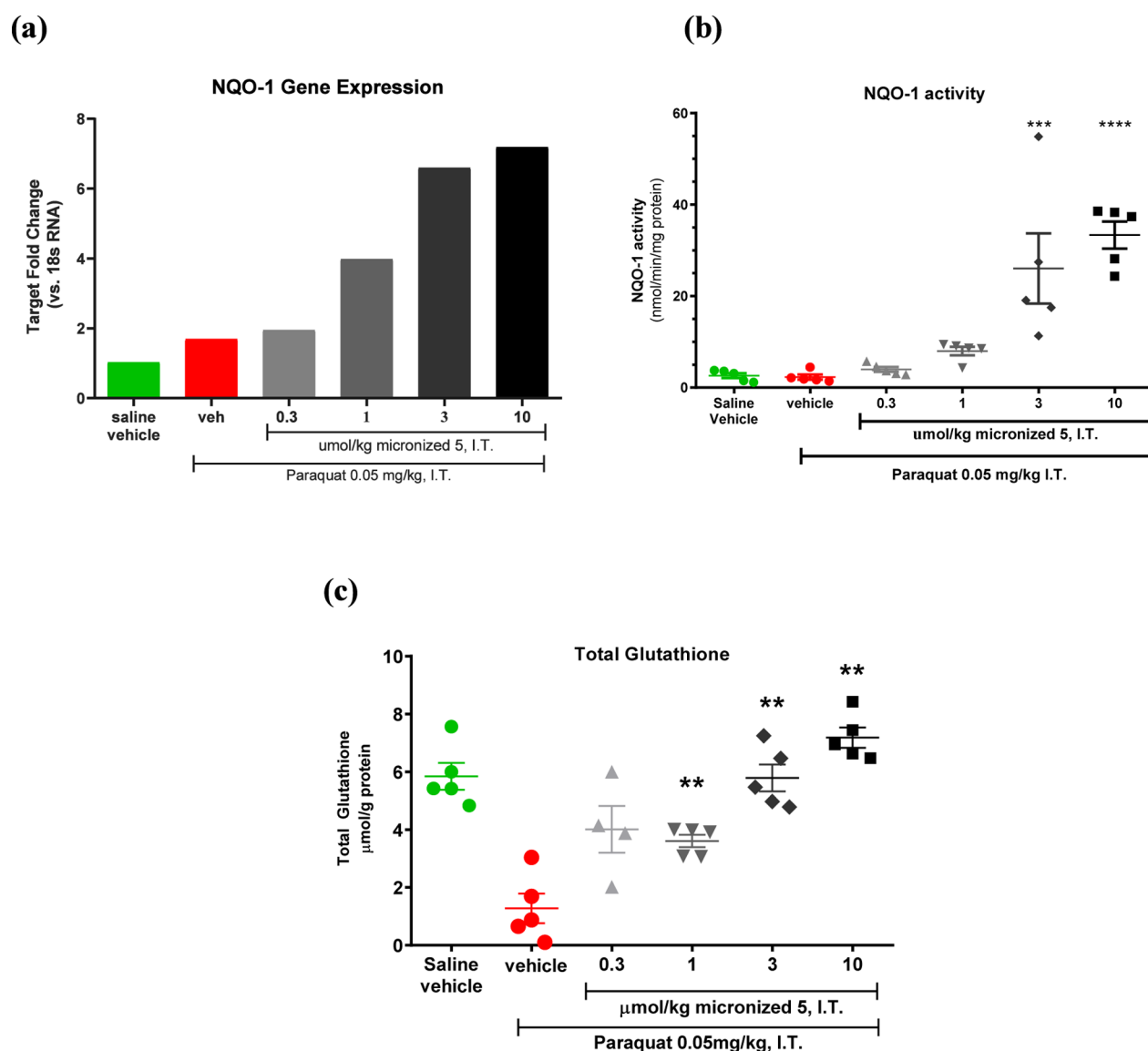


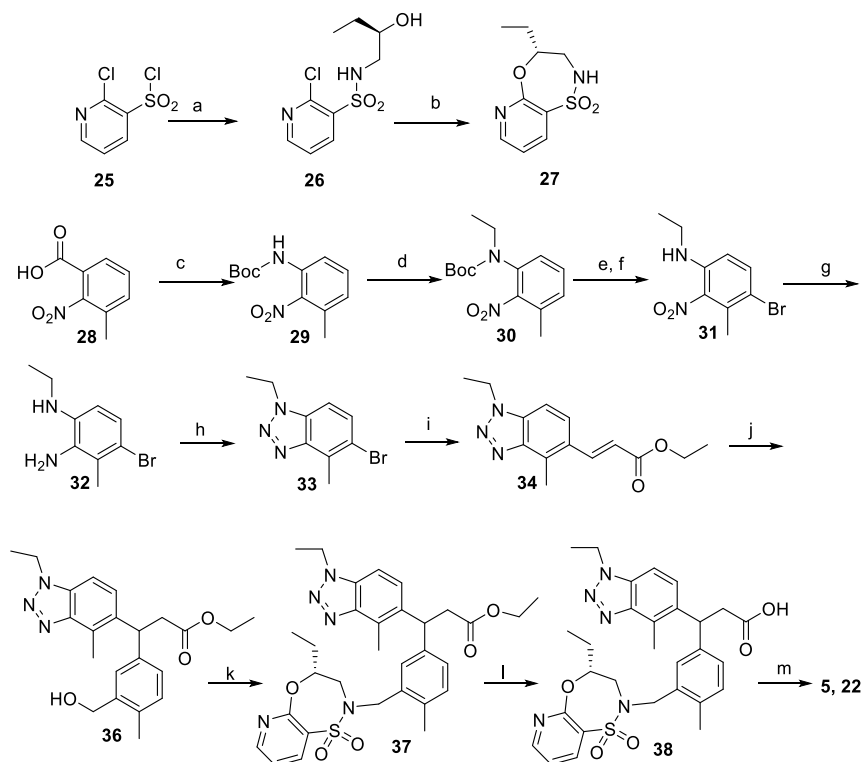
Figure 7. Intratracheal delivery of a wet suspension of micronized **5** (0.3–10 $\mu\text{mol/kg}$), 24 h prior to challenge with PQ (0.05 mg/mL given IT, approximate EC_{80}), lung homogenate evaluated 48 h post PQ challenge. Target engagement was demonstrated by a dose-dependent increase in NQO1 gene expression (a) and NQO1 enzyme activity (b) in the lung. Efficacy was demonstrated by a corresponding increase in total lung glutathione (c). Statistical analysis: (b) one way ANOVA w/Dunnett's correction and *** $p < 0.001$, **** $p < 0.0001$; (c) students t -test unpaired ** $p < 0.01$.

kidney, NQO1 expression reached a maximal response at 10 $\mu\text{mol/kg}$, while the expression levels of TXNRD1 and SRXN1 reached a maximum at 5 $\mu\text{mol/kg}$, similar to that observed in the lung (Figure 6). Although the IV route of administration is not viable for chronic diseases like COPD and is not optimal for bolus injections, it does suggest a spectrum of acute indications that could be treated systemically using IV infusion delivery with a NRF2 pathway activator like **5**.

In Vivo Pharmacological Profiling of **5**: Intratracheal Administration

Considering that the target organ in COPD as well as other chronic respiratory diseases is the lung, we next explored target engagement by intratracheal administration of NRF2 activators to Lewis rats. Compound **5** displayed properties favorable to delivery by inhalation, including picomolar target affinity, high efficacy in lung epithelial cells, high systemic clearance, low absorption, and limited aqueous solubility (30 $\mu\text{g/mL}$ as

determined using simulated lung fluid). An initial proof of concept experiment utilized a model of acute lung injury (ALI) and vascular leak driven by mitochondrial oxidative stress.¹² This was achieved in this model by directly instilling paraquat (PQ or N,N' -dimethyl-4,4'-bipyridinium dihydrochloride) into the trachea of Lewis rats. Rats were treated with micronized **5** via intratracheal administration 24 h prior to a PQ challenge, and target engagement and efficacy were measured 48 h after the PQ challenge. Both NQO1 expression (Figure 7a) and protein levels (as measured by NQO1 enzyme activity) in the lung (Figure 7b) increased in a dose-dependent manner from 0.3 to 10 $\mu\text{mol/kg}$.¹² This increase in NRF2 pathway activation was also well correlated with a dose-dependent increase in total lung glutathione levels (Figure 7c). In subsequent studies, **5** has also demonstrated efficacy in both acute and chronic preclinical lung models of oxidative stress (with a variety of oxidative challenges including cigarette smoke and ozone) when delivered directly to the lung.¹²

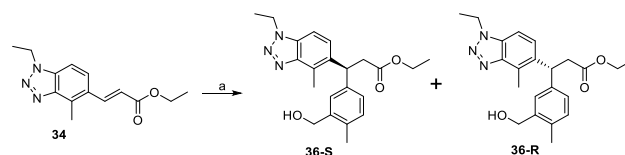
Scheme 1^a

^a(a) (R)-1-aminobutan-2-ol, K_2CO_3 , THF/ H_2O , 0 °C then room temperature, 16 h, 59%. (b) KOtBu, DMSO, 80 °C, 6 h, 59.9%; (c) diphenyl phosphorazidate, Et_3N , *t*-BuOH, 82 °C, 10 h, 90%; (d) NaH, DMF, 0 °C, 0.5 h; EtI, DMF, RT, 2 h, 83%; (e) TFA, CH_2Cl_2 , 30 °C, 16 h, 87%; (f) NBS, DMF, 0 °C, 0.3 h, 81%; (g) $SnCl_4 \cdot 2H_2O$, EtOH, RT; 75 °C, 6 h, 90%; (h) H_2SO_4 , H_2O , $NaNO_2$, 0 °C, 2 h, 95%; (i) tri-*o*-tolylphosphine, ethyl acrylate, $PdOAc_2$, K_2CO_3 , DMF, 120 °C, 12 h, 61.9%; (j) (3-(hydroxymethyl)-4-methylphenyl)boronic acid **35**, chloro(1,5-cyclooctadiene)rhodium(I) dimer, Et_3N , 1,4-dioxane, H_2O , 23 °C, 3 h, 98%; (k) **27**, nBu_3P , (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone), THF, 3 h, 99%; (l) LiOH, THF, H_2O , MeOH, 2 h, 84%; (m) chiral SFC, Chiralpak AD.

Synthesis

The preparation of **5** and **22** is outlined in Scheme 1. 2-Chloropyridine-3-sulfonyl chloride **25** was treated with (R)-1-aminobutan-2-ol in the presence of K_2CO_3 to provide the intermediate alcohol **26**, which was then cyclized by heating with KOtBu in DMSO to provide the cyclic sulfonamide **27**. Bromobenzotriazole **33** was prepared in six steps from 3-methyl-2-nitrobenzoic acid **28**. Palladium coupling of **33** to ethyl acrylate provided unsaturated ester **34**. Heating **34** and (2-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **35** with chloro(1,5-cyclooctadiene)rhodium(I) dimer in the presence of Et_3N provided the racemic bis-aryl alcohol **36**. Mitsunobu coupling of **36** with **27** using (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone) and nBu_3P in THF gave the intermediate ethyl ester **37** that was hydrolyzed with aqueous LiOH in MeOH to give the corresponding carboxylic acid **38**, which was subsequently separated by chiral SFC chromatography to provide diastereomerically pure **5** and **22**. The absolute stereochemistry of **5** and **22** was assigned by vibrational circular dichroism (VCD) analysis, and the absolute stereochemistry of **5** was later confirmed by small-molecule X-ray crystallography and protein X-ray crystallography of **5** bound to the Kelch domain of KEAP1. Details for the preparation of **7–21**, **23**, and **24** can be found in the Supporting Information.

As the team focused on profiling **5**, an enantiomerically enriched synthesis of the intermediate bisaryl alcohol **36-S** was developed (Scheme 2). Treatment of the unsaturated benzotriazole ester **34** and the boronic ester **35** with

Scheme 2^a

^a(a) 2-Methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **39**, bis(norbornadiene) rhodium(I) tetrafluoroborate, (R)-(+)-2,2'-bis(di-*p*-tolylphosphino)-1,1'-binaphthyl, 1,4-dioxane, KOH, H_2O , 25 °C, 20 h, 74.8%, ratio **36-S** to **36-R**: 82:16.

bis(norbornadiene)rhodium(I) tetrafluoroborate and (R)-(+)-2,2'-bis(di-*p*-tolylphosphino)-1,1'-binaphthyl in the presence of KOH in water and 1,4-dioxane at room temperature gave a mixture of **36-S** to **36-R** in the ratio of 82:16 that provided an enriched amount of the desired **36-S** required for the synthesis of **5**. Details for the preparation of **7–21** and **23–24** can be found in the Supporting Information.

DISCUSSION AND CONCLUSIONS

We have previously reported the application of fragment-based drug discovery to the identification of a novel series of bis-aryl PPI inhibitors that directly disrupt the interaction between NRF2 and KEAP1, resulting in the activation of the NRF2 pathway. Initial optimization of this series led to an early tool compound **4** that displayed tight binding to the Kelch domain and demonstrated proof of concept for NRF2 pathway

activation both in human cells and preclinically in vivo.^{9,10} This report builds on that earlier work to describe the discovery of a series of analogues with improved cellular efficacy and physicochemical properties, ultimately resulting in the discovery of GSK3227634, compound **5**. In addition to displaying the ability to activate the NRF2 pathway after systemic (IV infusion) administration, **5**, both engages the NRF2 pathway and demonstrates efficacy in the lung following intratracheal administration. Although there are other examples of potent noncovalent KEAP1-NRF2 inhibitors (e.g., PRL-295 and CUPY192018),⁷ compound **5** described here has the appropriate balance of target affinity, cellular efficacy, and pharmacokinetic and physicochemical properties likely needed to display significant in vivo preclinical target engagement and efficacy after inhaled delivery. Although intratracheal administration does not precisely replicate inhalation-associated drug distribution in the human lung, it provides a close approximation for early experimental studies in rodents. Inhaled administration of an NRF2 activator provides several advantages when treating COPD and other chronic lung diseases, including increased efficacy due to direct delivery to the target organ, the potential to limit adverse systemic events by limiting systemic exposure, and the potential to combine with therapeutics targeting other mechanisms of action, e.g., bronchodilators. The latter can be important when targeting multifaceted chronic diseases such as COPD.

In our previous structure-driven studies on the bis-aryl series, we noted that most of the polar and π -stacking features in the NRF2-binding pocket were well satisfied by elaborated ligands, necessitating approaches other than optimizing noncovalent interactions to further enhance affinity and cellular efficacy.^{9,10} Increasing ligand preorganization is an important route to improving affinity,¹⁷ and may be particularly beneficial in the context of inhibiting protein–protein interactions, allowing potency gains to be achieved without addition of significant molecular weight. We previously demonstrated that the specific addition of a methyl group to the oxathiazepine in **4** was associated with an increased population of the bioactive conformation, through a combination of local conformational restriction and “hydrophobic collapse”. Based on this observation and mindful of the requirement of high-affinity binding for low-dose inhaled delivery, we sought to further increase the propensity for the bound conformation during lead optimization through homologation of methyl to ethyl in both benzotriazole and oxathiazepine rings. The solution-phase NMR and small-molecule X-ray crystallographic data reported here support the hypothesis that free **5** adopts a major conformation, which is very similar to the conformation bound to KEAP1 and may contribute to its very high affinity and excellent cellular efficacy. In addition, our bis-aryl compounds have been shown to be very selective activators of the NRF2 pathway, and this has been further demonstrated for **5** by extensive selectivity and proteomic profiling. This is in contrast to the reversibly covalent compounds targeting reactive cysteine residues in the BTB/IVR domains of KEAP1, which can be more promiscuous in their interactions.¹⁸ This increased target selectivity, in combination with inhaled delivery that limits systemic exposure, provides a much more selective activator of the NRF2 pathway, which may be beneficial for the treatment of chronic lung diseases including COPD.

In summary, we have discovered **5**, a selective and ultrahigh-affinity inhibitor of the binding of NRF2 to the KEAP1 Kelch domain that activates the NRF2 pathway both in vitro in human

lung epithelial cells as well as in vivo after intratracheal administration in a rat model of lung oxidative stress.

■ EXPERIMENTAL PROCEDURES

Preparation of Compounds **5** and **22**

All solvents and commercially available reagents were used as received. All reactions were followed by thin-layer chromatography (TLC) analysis (TLC plates GF254, Merck) or liquid chromatography mass spectrometry (LC–MS). Intermediates and final products were purified by either flash column chromatography or semiprep reverse-phase HPLC. Flash chromatography was performed on columns prepacked with silica gel from Teledyne ISCO, unless otherwise noted. Analytical LC–MS analysis was completed on a Sciex 7 or Waters 9 LC–MS system. Preparative reverse-phase HPLC was performed using a Gilson Preparative System with variable wavelength UV detection or an Agilent Mass Directed AutoPrep (MDAP) system with both mass and variable wavelength UV detection or Waters Preparative System with UV/PDA detection or a Shimadzu PREP LC 20AP. A variety of reverse phase columns, e.g., Luna 5 μ C18(2) 100A, SunFire C18, XBridge C18, and Atlantis T3, were used in the purification, with the choice of column support dependent upon the conditions used in the purification. The compounds are eluted using a gradient of CH₃CN and water. Neutral conditions used a CH₃CN and water gradient with no additional modifier, acidic conditions used an acid modifier, 0.1% TFA (added to both the CH₃CN and water) or 0.1% formic acid and basic conditions used a basic modifier, 0.1% NH₄OH (added to the water) or 10 mM ammonium bicarbonate. Preparative Chiral SFC was performed using a Thar/Waters Preparative SFC System with a single-wavelength UV detection system or PDA detector. A variety of chiral SFC columns (e.g., ChiralPak IA, IC, AY, AD, OD, OJ, and C2) were used in the purification. The compounds are eluted using supercritical fluid CO₂ and cosolvents such as MeOH, EtOH, IPA, or a combination of these solvents in different ratios. Modifiers (e.g., 0.1% TFA, NH₄OH, and DEA) were used as needed. Routine proton nuclear magnetic resonance (¹H NMR) spectra were recorded with a Bruker NMR spectrometer unless otherwise noted in the indicated solvent. NMR spectra are referenced as follows: ¹H (400 or 500 MHz), internal standard TMS at δ = 0.00. Abbreviations for multiplicities observed in NMR spectra: s, singlet; br s, broad singlet; d, doublet; t, triplet; q, quadruplet; p, pentuplet; spt, septuplet; m, multiplet. IR and VCD spectra were recorded on a ChiralIR2XTM VCD spectrometer (BioTools, Inc.) equipped with a dual PEM accessory (additional details around the methodology used can be found in the preparation section of the [Supporting Information](#)). All final compounds reported are of at least 95% purity according to LC–MS unless stated otherwise.

(R)-2-Chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide (26). To a solution of 2-chloropyridine-3-sulfonyl chloride **25** (30 g, 141 mmol) in a mixture of tetrahydrofuran (200 mL) and water (100 mL) was added (R)-1-aminobutan-2-ol (18.92 g, 212 mmol) and K₂CO₃ (23.46 g, 170 mmol) at 0 °C. The reaction mixture was stirred at room temperature for 16 h. The reaction mixture was evaporated under reduced pressure, quenched with water, extracted twice with DCM, and the organic layer was dried with anhydrous Na₂SO₄ and filtered. The filtrate was evaporated under reduced pressure to afford (R)-2-chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide **26** (28 g, 84 mmol, 59.0% yield) as a pale yellow liquid that was used in the next reaction without any further purification. LC–MS: m/z : 265.02 [M + H]⁺, 1.68 min (ret. time); ¹H NMR (400 MHz, chloroform-*d*): δ = 8.58 (dd, *J* = 1.8, 4.8 Hz, 1H), 8.41 (dd, *J* = 1.9, 7.8 Hz, 1H), 7.43 (dd, *J* = 4.7, 7.8 Hz, 1H), 3.64 (br dd, *J* = 2.9, 5.5 Hz, 1H), 3.13 (dd, *J* = 3.2, 13.0 Hz, 1H), 2.84 (dd, *J* = 8.0, 13.0 Hz, 1H), 1.58–1.38 (m, 2H), 1.02–0.84 (m, 3H).

(R)-4-Ethyl-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]-oxathiazepine 1,1-Dioxide (27). To a solution of (R)-2-chloro-N-(2-hydroxybutyl)pyridine-3-sulfonamide **26** (25 g, 74.6 mmol) in dimethyl sulfoxide (200 mL) was added KOtBu (16.74 g, 149 mmol) at 0 °C. The reaction mixture was stirred at 80 °C for 6 h. The reaction mixture was quenched with ice water (150 mL), neutralized with 1 N HCl (up to pH = 6), extracted with EtOAc (3 $\times$ 300 mL), and the organic layer was washed with cold water (3 $\times$ 300 mL), brine (1 $\times$ 200

mL), and dried with anhydrous Na_2SO_4 and filtered. The filtrate was evaporated under reduced pressure to get crude compound. The crude compound was purified by column chromatography using EtOAc:hexane (34:66) as a solvent. The eluted fractions were concentrated under vacuum to obtain the compound, which was washed with petroleum ether (30 mL) + diethyl ether (15 mL) to afford (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepine 1,1-dioxide **27** (8.8 g, 38.6 mmol, 51.7% yield) as a white solid. This material was blended with a previous preparation, and the blended sample was washed with pentane (30 mL) + diethyl ether (10 mL) to afford **27** (10.2 g, 44.7 mmol, 59.9% yield) as a white solid. LC–MS m/z : 229.16 $[\text{M} + \text{H}]^+$, 1.38 min (ret. time); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 8.46 (dd, J = 2.0, 4.8 Hz, 1H), 8.17 (dd, J = 1.9, 7.6 Hz, 1H), 8.01 (s, 1H), 7.38 (dd, J = 4.8, 7.7 Hz, 1H), 3.98–3.85 (m, 1H), 3.51–3.34 (m, 2H), 1.70–1.56 (m, 2H), 1.04 (t, J = 7.3 Hz, 3H).

tert-Butyl (3-Methyl-2-nitrophenyl)carbamate (29). To the mixture of 3-methyl-2-nitrobenzoic acid **28** (50 g, 276 mmol) in *tert*-butanol (250 mL) was added diphenyl phosphorazidate (84 g, 304 mmol) and TEA (42.3 mL, 304 mmol). The mixture was stirred at 82 °C for 10 h, and then concentrated in vacuo. The residue was dissolved in EtOAc, washed with aqueous NaHCO_3 , dried over Na_2SO_4 , and concentrated in vacuo to give *tert*-butyl (3-methyl-2-nitrophenyl)-carbamate **29** (67 g, 248 mmol, 90% yield). LC–MS m/z : 153 $[\text{M} + \text{H}-100]^+$, 1.95 min (ret. time).

tert-Butyl Ethyl(3-methyl-2-nitrophenyl)carbamate (30). To a solution of *tert*-butyl (3-methyl-2-nitrophenyl)carbamate **29** (35 g, 139 mmol) in *N,N*-dimethylformamide (400 mL) was added NaH (6.10 g, 153 mmol) slowly under nitrogen at 0 °C. The reaction mixture was stirred at 0 °C for 0.5 h. An iodoethane (23.80 g, 153 mmol) solution in DMF (100 mL) was added dropwise and was stirred at room temperature for 2 h. It was added to water (500 mL) and extracted with ethyl acetate (300 mL $\times$ 3). Isolate was added to the combined organic layers, and the mixture was concentrated under a stream of nitrogen at 50 °C. *tert*-Butyl ethyl(3-methyl-2-nitrophenyl)carbamate **30** (36 g, 116 mmol, 83% yield) was obtained, which was used in the next step without further purification. LC–MS m/z : 303.1 $[\text{M} + 23]^+$, 1.29 min (ret. time).

4-Bromo-N-ethyl-3-methyl-2-nitroaniline (31). A solution of *tert*-butyl ethyl (3-methyl-2-nitrophenyl)carbamate **30** (5 g, 17.84 mmol) in dichloromethane (20 mL) was added TFA (6 mL, 78 mmol) slowly under nitrogen at 30 °C. The reaction mixture was stirred at 30 °C for 16 h. The reaction mixture was concentrated to remove the DCM and TFA. The pH was adjusted to 9 with NH_4OH and then extracted with ethyl acetate (30 mL $\times$ 3) to obtain after evaporation *N*-ethyl-3-methyl-2-nitroaniline (3 g, 15.48 mmol, 87% yield), which was used in the next step without further purification. LC–MS m/z : 181.1 $[\text{M} + \text{H}]^+$, 1.25 min (ret. time). A solution of *N*-ethyl-3-methyl-2-nitroaniline (3.5 g, 18.06 mmol) in *N,N*-dimethylformamide (80 mL) was added to NBS (3.21 g, 18.06 mmol) solution in *N,N*-dimethylformamide (80 mL) slowly under nitrogen at 0 °C. The mixture reaction was stirred at 0 °C for 0.3 h. The reaction mixture was added to water (200 mL) and extracted with ethyl acetate (50 mL $\times$ 3). Isolate was added to the combined organic layers, and the mixture was concentrated under a stream of nitrogen at 50 °C. 4-Bromo-*N*-ethyl-3-methyl-2-nitroaniline **31** (4.2 g, 14.59 mmol, 81% yield) was obtained and was used in the next step without further purification. LC–MS m/z : 261.0 $[\text{M} + \text{H}]^+$, 1.33 min (ret. time).

4-Bromo-N1-ethyl-3-methylbenzene-1,2-diamine (32). A solution of 4-bromo-*N*-ethyl-3-methyl-2-nitroaniline **31** (4.2 g, 16.21 mmol) in ethanol (20 mL) was added to tin(II) chloride dihydrate (9.14 g, 40.5 mmol) slowly under nitrogen at room temperature. The reaction mixture was stirred at 75 °C for 6 h. The reaction mixture was concentrated to remove most of the ethanol. Water (20 mL) was added, cooled to 0 °C, and the pH was adjusted to 14 with 6 N NaOH. It was extracted with ethyl acetate (50 mL $\times$ 3), and the ethyl acetate extracts were concentrated. The Isolute-adsorbed crude product was added to a silica gel column and was eluted with hexane/ethyl acetate = 4:1 to give 4-bromo-*N*1-ethyl-3-methylbenzene-1,2-diamine **32** (3.5 g, 14.51 mmol, 90% yield), which was used in the next step without further purification. LC–MS m/z : 229.1 $[\text{M} + \text{H}]^+$, 0.99 min (ret. time).

5-Bromo-1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazole (33).

A solution of 4-bromo-*N*-1-ethyl-3-methylbenzene-1,2-diamine **32** (3.5 g, 15.28 mmol) in H_2SO_4 (2.9 mL, 54.4 mmol) solution in water (30 mL) was slowly added a sodium nitrate (1.948 g, 22.91 mmol) solution in water (5 mL) under nitrogen at 0 °C. The reaction mixture was stirred at 0 °C for 2 h. The reaction mixture was filtered, and the solid was dried under high vacuum to give 5-bromo-1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazole **33** (3.5 g, 14.58 mmol, 95% yield). LC–MS m/z : 240.1 $[\text{M} + \text{H}]^+$, 1.71 min (ret. time).

(E)-Ethyl 3-(1-Ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate (34). A mixture of tri-*o*-tolylphosphine (0.444 g, 1.458 mmol), ethyl acrylate (2.92 g, 29.2 mmol), 5-bromo-1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazole **33** (3.5 g, 14.58 mmol), PdOAc_2 (0.164 g, 0.729 mmol), K_2CO_3 (4.03 g, 29.2 mmol), and *N,N*-dimethylformamide (200 mL) was stirred at 120 °C for 12 h. The mixture was poured into water and extracted with ethyl acetate (30 mL $\times$ 3). The organic layer was dried and concentrated. The Isolute-adsorbed crude product was added to a silica gel column and was eluted with hexane/ethyl acetate = 4:1 to give (E)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **34** (2.34 g, 9.02 mmol, 61.9% yield) as a solid. LC–MS m/z : 260.1 $[\text{M} + \text{H}]^+$, 1.67 min (ret. time); ^1H NMR (400 MHz, chloroform-*d*): δ = 8.16 (d, J = 16.1 Hz, 1H), 7.72 (d, J = 8.8 Hz, 1H), 7.37 (d, J = 8.8 Hz, 1H), 6.43 (d, J = 16.1 Hz, 1H), 4.69 (q, J = 7.3 Hz, 2H), 4.31 (q, J = 7.0 Hz, 2H), 2.93 (s, 3H), 1.68–1.61 (m, 3H), 1.38 (t, J = 7.2 Hz, 3H).

Ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (36). (E)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **34** (1.037 g, 4.0 mmol), (3-(hydroxymethyl)-4-methylphenyl)boronic acid **35** (0.996 g, 6.00 mmol), chloro (1,5-cyclooctadiene)rhodium(I) dimer (0.099 g, 0.200 mmol), and triethylamine (0.558 mL, 4.00 mmol) were added to 1,4-dioxane (28 mL) and water (14 mL) in a Schlenk flask, and a stream of argon was bubbled through the mix for ~2 min. The reaction was stirred under argon for 3 h at 23 °C. Solvent was removed in vacuo, and the residue was diluted with DCM (15 mL), filtered through a glass fiber filter, and then passed through a hydrophobic frit. The filtrate was pre-absorbed on Isolute, and the crude product was purified on a silica cartridge (80 g) with a Combiflash Companion, eluting at 60 mL/min with a gradient running from DCM to 60% EtOAc in DCM over 30 min. The desired fractions were pooled and concentrated to afford ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36** (1.50 g, 3.93 mmol, 98% yield) as a gummy yellow foam. LC–MS m/z : 382.3 $[\text{M} + \text{H}]^+$, 0.92 min (ret. time); ^1H NMR (400 MHz, chloroform-*d*): δ = 7.40–7.28 (m, 3H), 7.10 (s, 2H), 5.01 (t, J = 8.0 Hz, 1H), 4.70–4.63 (m, 4H), 4.04 (q, J = 7.2 Hz, 2H), 3.20–3.03 (m, 2H), 2.88 (s, 2H), 2.31 (s, 3H), 1.64–1.58 (m, 5H), 1.13 (t, J = 7.2 Hz, 3H).

(S)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid (5) and (R)-3-(3-(((R)-4-Ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic Acid (22). Ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36** (1.5 g, 3.93 mmol), (R)-4-ethyl-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepine 1,1-dioxide **27** (0.942 g, 4.13 mmol), and (E)-diazene-1,2-diylbis(piperidin-1-ylmethanone) (1.984 g, 7.86 mmol) were dissolved in tetrahydrofuran (42 mL), and tributylphosphine (1.964 mL, 7.86 mmol) was added. The reaction mixture was stirred in a round-bottom flask under Ar for 3 h. The solvent was removed in vacuo, redissolved in DCM, and pre-absorbed on Isolute. The crude product was purified on a silica cartridge (120 g) with a Combiflash Companion, eluting at 85 mL/min with a gradient running from hexanes to EtOAc over 30 min. The desired fractions were pooled and concentrated to afford ethyl 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate **37** (2.3 g, 3.89 mmol, 99% yield). LC–MS m/z : 592.4 $[\text{M} + \text{H}]^+$, 1.12 min (ret. time).

Ethyl-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoate **37** (2.3 g, 3.89 mmol) was dissolved in tetrahydrofuran (30 mL), and a solution of lithium hydroxide (1.862 g, 78 mmol) dissolved in water (30.0 mL) was added. Methanol (50 mL) was added, sufficient to form a cloudy solution, and the mixture was stirred 2 h. The volatile solvent was removed in vacuo, and the residue was diluted with EtOAc (100 mL) and water (40 mL), and the phases were shaken (aqueous phase was pH 10 by pH paper). The basic phase extract was removed, and the aqueous phase was combined with 6 M HCl (20 mL, 120 mmol) extracted again with EtOAc (75 mL), and the acid phase EtOAc extract was dried (MgSO₄) and concentrated to afford 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **38** (1.84 g, 3.26 mmol, 84% yield). LC–MS *m/z* 564.2 [M + H]⁺, 0.95 min (ret. time); ¹H NMR (400 MHz, chloroform-*d*): δ = 8.50 (br s, 1H), 8.27 (d, *J* = 7.5 Hz, 1H), 7.37–7.30 (m, 2H), 7.15 (s, 2H), 7.07 (d, *J* = 7.3 Hz, 1H), 4.97 (br d, *J* = 3.8 Hz, 1H), 4.67 (q, *J* = 7.0 Hz, 2H), 4.49–4.34 (m, 2H), 4.09–4.01 (m, 1H), 3.58 (br s, 1H), 3.22–3.13 (m, 1H), 3.13–3.03 (m, 2H), 2.84 (s, 3H), 2.33 (s, 3H), 2.12 (s, 3H), 1.71–1.54 (m, 4H), 1.00–0.88 (m, 3H). Two batches of 3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **38** were combined and purified by chiral SFC: Chiralpak AD, 20 × 250 mm, 5 μ; cosolvent: 30% MeOH; flow rate: 50 mL/min; back pressure: 100 bar to provide the pure diastereoisomers, (S)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **5** and (R)-3-(3-(((R)-4-ethyl-1,1-dioxido-3,4-dihydro-2H-pyrido[2,3-*b*][1,4,5]oxathiazepin-2-yl)methyl)-4-methylphenyl)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)propanoic acid **22**. The absolute stereochemistry of **5** and **22** were assigned by VCD analysis. The absolute stereochemistry of **5** was also confirmed by small molecule X-ray crystallography as well as protein X-ray crystallography of **5** bound to the Kelch domain of KEAP1.

Alternate Synthesis of ethyl (S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (36-S) 2-Methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol (39). To a solution of (S)-bromo-2-methylphenylmethanol (30 g, 149 mmol) in 1,4-dioxane (200 mL) was added potassium acetate (43.9 g, 448 mmol) and bis(pinacolato)diboron (56.8 g, 224 mmol), and the reaction mixture was degassed with Ar for 30 min and then added PdCl₂(dppf)-CH₂Cl₂ adduct (7.31 g, 8.95 mmol) to reaction mixture and stirred at 100 °C for 16 h. After cooling to 25 °C, the mixture was filtered through a Celite pad, the filtrate was concentrated under vacuum, and the solvent was removed in vacuo. The residue was purified by Combiflash (silica gel column, 330 g, eluted with PE/EA = 4:1) to give (2-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **39** (33 g, 127 mmol, 85% yield) as a yellow oil. LC–MS *m/z* 231.1 [M + H-18]⁺, 1.92 min (ret. time); ¹H NMR (500 MHz, CDCl₃): δ 7.80 (d, *J* = 30.7 Hz, 1H), 7.64 (dd, *J* = 13.7, 7.5 Hz, 1H), 7.17 (dd, *J* = 20.0, 7.5 Hz, 1H), 4.93 (s, 1H), 4.70 (s, 1H), 2.36 (d, *J* = 25.1 Hz, 3H), 1.33 (d, *J* = 5.2 Hz, 12H).

Ethyl-(S)-3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (36-S). 2-Methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)methanol **39** (2.87 g, 11.57 mmol), (E)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)acrylate **34**, (1 g, 3.86 mmol), bis-(norbornadiene)rhodium(I) tetrafluoroborate (0.144 g, 0.386 mmol), and (R)-(+)-2,2'-bis(di-*p*-tolylphosphino)-1,1'-binaphthyl (0.283 g, 0.417 mmol) were combined in 1,4-dioxane (20 mL). To the mixture, a solution of potassium hydroxide (0.216 g, 3.86 mmol) in water (3.86 mL) was added. The mixture was stirred at 25 °C for 20 h under the protection of nitrogen. The reaction mixture was diluted with water (20 mL) and extracted with EtOAc (20 × 3 mL), dried, and concentrated. The product was purified by silica gel column (EtOAc/PE = 4:5). Ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-

3-(3-(hydroxymethyl)-4-methylphenyl)propanoate (1.1 g, 2.88 mmol, 74.8% yield) was obtained as a solid mixture of enantiomers. LC–MS: *m/z* 382.2 [M + H]⁺, 1.58 min (ret. time). Chiral HPLC analysis showed an **36-S:36-R** ratio of 82:16. This material was combined with another 4 g batch with about the same **36-S:36-R** ratio and purified by chiral SFC (Chiralpak AD-H, 50 × 250 mm, 5 μ (Daicel); column temperature: 35 °C; mobile phase: CO₂/methanol (0.5% DEA) = 60/40; flow: 160 g/min; back pressure: 100 bar) to obtain (S)-ethyl 3-(1-ethyl-4-methyl-1H-benzo[d][1,2,3]triazol-5-yl)-3-(3-(hydroxymethyl)-4-methylphenyl)propanoate **36-S** (3.8 g, 9.25 mmol) as solid. LC–MS *m/z*: 382.2 [M + H]⁺, 1.58 min (ret. time).

The detailed preparation of **7–21** and **23–24** can be found in the [Supporting Information](#).

Small Molecule X-ray Diffraction Study of Compound 5

Data were collected with an Oxford Diffraction Gemini A Ultra diffractometer at 150(2)K using Cu-Kα X-radiation (λ = 1.54178 Å). Crystal data and refinement summary for **5**: C₂₉H₃₃N₅O₅S; M = 563.66; colorless block from the slow evaporation of a solution of **5** in acetonitrile and water; 0.38 × 0.25 × 0.19 mm; trigonal; space group P3₂ (#145); *a* = 14.25220(10) Å, *c* = 12.04696(10) Å, *V* = 2119.20(3) Å³; *Z* = 3; *D*_{calc} = 1.325 Mg m⁻³; θ_{max} = 66.37°; reflections collected = 58,212; independent reflections = 4929; *R*_{int} = 0.0277; coverage = 99.9%; restraints = 1; parameters = 370; *S* = 1.042; *R*₁ [*I* > 2σ(*I*)] = 0.0227; *wR*₂ (all data) = 0.0610; and largest difference peak and hole = 0.224 and -0.182 e-Å⁻³. A description of the refinement and the full tables associated with the crystal structure are given in the [Supporting Information](#). Further details of the data collection and refinement can be found in the crystallographic information file (CIF) deposited as ESI, together with the check CIF output. The CIF has also been deposited with the Cambridge Crystallographic Data Centre (CCDC 2541002). The data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/structures.

Crystal Structure of the Kelch Domain of KEAP1 in Complex with Compound 5

As described in our previous studies, we used mouse KEAP1 as a surrogate for the human orthologue for protein–ligand crystallography. The amino acid sequence identity between murine and human Kelch domains is 97%, and the nine residues that are different are mostly conservative replacements, with none in the ligand binding site. Murine KEAP1 (325–613) protein containing a His₆ tag at the N-terminus followed by a thrombin cleavage site was purified as described previously.⁶ The protein was concentrated to 17 mg/mL in 50 mM Tris, pH 8.0, 20 mM DTT, 10 mM benzamidine, and stored at -80 °C. The protein was crystallized by mixing protein and reservoir solution at a 2:1 ratio (protein: reservoir) at 22 °C in Hampton Cryschem M 24-well sitting drop plates. The droplets were equilibrated against 300 μL of reservoir solution containing 0.9 M lithium sulfate, 0.45 M ammonium sulfate, 0.1 M sodium citrate pH 5.6 and 0.2 M NDSB-221. For soaking the compound into the crystals, a Hampton Cryschem M 24-well sitting drop plate was prepared by filling the reservoir compartment with 500 μL of 25% PEG4000 and 0.1 M Bis-Tris, pH 7.0. The crystals were harvested and soaked in 3–5 μL of the reservoir solution containing the ligand at 0.25–1 mM (final concentration) drawn from a stock of 100 mM 100% DMSO. The crystals were incubated for up to 24 h before harvesting into the soaking solution containing 6.25–25% ethylene glycol and plunged in liquid nitrogen.

X-ray diffraction data were collected at APS SER-CAT. The diffraction spots were integrated and scaled with HKL2000 (see [Table S1](#)).¹⁹ The structure was determined by molecular replacement with Phaser²⁰ using the 1U6D structure without ligands or waters as the search model. The model was refined with Phenix refine²¹ and manually adjusted using COOT.²² The final model includes residues 325 to 614 ([Table S1](#)).

NMR Conformational Studies of Compound 5

NMR spectra (¹H, HSQC, and ROESY) for GSK3227634 (**5**) were acquired in sodium phosphate buffer (pH 7.1) at 296 K and DMSO-*d*₆ at 300 K on a Bruker Avance III 500 MHz spectrometer with a 5 mm QCI cryoprobe using 3 mm capillary tubes with a sample volume of 160

μL (Bruker). COSY and HMBC spectra were also acquired for the sample in $\text{DMSO-}d_6$. Data processing was performed using MestReNova 14 (Mestre Lab Research, Santiago de Compostela, Spain). The sample in sodium phosphate buffer was prepared from a 10 mM stock in $\text{DMSO-}d_6$ and diluted 20-fold with 50 mM sodium phosphate buffer in D_2O (pH 7.1) to afford a nominal final concentration of 0.5 mM. **5** was soluble at this concentration as determined by quantitative NMR (QUANTAS approach²³) using the qNMR plugin in MestReNova 14. The ^1H NMR spectrum in phosphate buffer with presaturation (1.6 s) of residual HOD was acquired with 32K (real + imaginary) data points. The real FID was zero-filled to 64K and Fourier transformed with an exponential line broadening of 0.3 Hz. The resulting spectrum was baseline-corrected (Whittaker Smoother) and referenced to the DMSO resonance at 2.69 ppm that corresponds to TSP = 0 ppm. The ROESY spectrum with presaturation (2 s) of residual HOD and mixing time (250 ms) was acquired with a sweep width of 12 ppm with 2K (real + imaginary) data points in f_2 and 256 (real + imaginary) increments in f_1 . The f_2 dimension was zero-filled to afford 2048 real data points. Linear prediction was applied to the f_1 dimension to afford 256 real data points that were also zero-filled to 2048. Both dimensions were Fourier transformed with a 90° sine squared window function and baseline-corrected. %ROE measurements were taken from integration of slices in f_2 at the f_1 chemical shift of interest. The peak area of the hydrogen of interest was scaled to -100% for a CH and -300% for a CH_3 .

Kelch NRF2 Fluorescence Polarization Assay

Inhibition of the Kelch domain-NRF2 interaction was determined using a fluorescence polarization-based competition assay in black 384-well microplates containing compounds of interest. Each assay well contained 11 nM 5'-TAMRA-NRF2 ETGE motif-containing peptide (AFFAQLQLDEETGEFL, 21st Century Biochemicals) and 25 nM human KEAP1 Kelch domain (321–609) protein in 10 μL assay buffer (50 mM Tris, pH 8.0, 100 mM NaCl, 5 mM MgCl_2 , 1 mM DTT, 2 mM CHAPS, and 0.005% BSA). Fluorescence polarization (excitation 540 nm/emission 590 nm) was measured using a BMG Pherastar FS plate reader. The pIC_{50} values of compounds on inhibition of Kelch-NRF2 interaction were determined by 4-parameter fitting of 11-points inhibition dose–response curves using ActivityBase (IDBS, Boston MA). pIC_{50} shown is the arithmetic mean of independent determinations.

NRF2-KEAP1 TR-FRET Low Protein Assay

In the NRF2-KEAP1 TR-FRET LP assay, full-length NRF2 protein and full-length KEAP1 protein (KEAP1 exists as a dimer) were used. The assay detects a compound's ability to inhibit the binding of KEAP1-FlagHis protein by biotinylated Avi-NRF2 protein. Assays were performed in black 384-well microplates with compounds of interest predispensed. Compounds were first incubated at 37°C for 2.5 h with 1.7 nM KEAP1-FlagHis protein in 5 μL assay buffer (50 mM Tris, pH 8.0, 5 mM MgCl_2 , 100 mM NaCl, 0.005% BSA, 1 mM DTT, and 2 mM CHAPS). Five microliter of 3.4 nM biotin-NRF2 protein in assay buffer was then added per well, followed by a 1 h incubation at 4°C . Ten microliter of detection mix (1 nM Streptavidin Eu + W1024 and 5 $\mu\text{g}/\text{mL}$ mouse anti-DYKDDDDK IgG conjugated to SureLight APC antibody; both from Columbia Biosciences, in assay buffer) was then added to each well. After 1 h incubation at room temperature, the assay plates were read on an Envision plate reader using a 320 nm excitation filter and 615 and 665 nm emission filters. The pIC_{50} values of compounds on inhibition of KEAP1-NRF2 interaction were determined by 4-parameter fitting of 11-point inhibition dose–response curves using ActivityBase (IDBS, Boston, MA). pIC_{50} shown is the arithmetic mean of independent determinations.

BEAS-2B NQO1 MTT Assay

BEAS-2B cells (ATCC) were plated at 50 μL per well in 384-well black clear-bottomed plates and were incubated at 37°C , 5% CO_2 overnight. On day two, the plates were centrifuged, and 50 nL of compound or controls was added to the cells. Plates were then incubated at 37°C , 5% CO_2 for 48 h. On day four, the medium was removed from the plates, and crude cell lysates were made by adding 13 μL per well of 1X lysis

buffer (Cell Signaling Technology, Boston MA) with Complete, Mini, EDTA-free Protease Inhibitor (Roche Applied Sciences, Germany). The plates were incubated for 20 min at room temperature for cell lysis, and 50 μL MTT cocktail was added to each well for the measurement of NQO1 activity.²⁴ The samples were analyzed on an Envision plate reader (PerkinElmer), reading absorbance at 570 nm for 30 min. Product formation was measured kinetically, and the pEC_{50} of NQO1 induction by compounds were calculated by plotting the change in absorbance ($\Delta\text{OD}/\text{min}$) versus $\log[\text{compound}]$ followed by 3-parameter fitting with ActivityBase (IDBS, Boston, MA). pEC_{50} shown is the arithmetic mean of independent determinations.

Artificial Membrane Permeability Assay

The artificial phospholipid membrane technique is similar to the widely used Caco-2 cell monolayer permeation method. Egg phosphatidylcholine (1.8%) and cholesterol (1%) were dissolved in *n*-decane. A small amount of this mixture was applied to the bottom of the microfiltration filter insert. Phosphate buffer (pH 7.4) was quickly added to the donors and receivers, and the lipids were allowed to form self-assembled lipid bilayers across the small holes in the filter. Permeation experiments were initiated by spiking the compounds of interest to the donor sides, and the experiment was stopped at a predetermined elapsed time. The samples were withdrawn and transferred to vials for analysis by HPLC with UV detection.

Surface Plasmon Resonance K_d Determination for Compound 5

SPR experiments were conducted on a Biacore T200 instrument (Cytiva). The Kelch domain of murine KEAP1 (322–624) was biotinylated during purification using an modular intein ligation approach²⁵ and captured on three flow cells of a streptavidin sensor chip, and one flow cell was reserved as a reference. Binding of compound **5** (TFA salt) was assessed at 25°C in 10 mM HEPES buffer pH 7.4, 150 mM NaCl with 0.005% P20, 1% DMSO, and 0.5 mM TCEP. Analyte was injected using a Biacore single-cycle kinetic method over all four flow cells with a 5-point 3-fold dilution series and top concentration of 100 nM. Measurements were carried out with a 60 s contact time and 1000 and 14,000 s dissociation times for the NRF2 peptide and compound **5**, respectively. Referenced data were obtained by subtracting the response from the protein-free reference flow cell from the response of the flow cells containing KEAP1. Subsequently, double referencing was achieved by subtracting a series of buffer-blank injections from the referenced data. Double referenced data were fit to a 1:1 kinetic binding model to obtain the binding affinity, K_d , which was determined as the ratio $k_{\text{off}}/k_{\text{on}}$, and $t_{1/2} = \ln(2)/k_{\text{off}}$.

ChromlogD Determinations

ChromlogD_{7.4} determinations were carried out at pH 7.4 utilizing literature protocols.²⁶

MDCK Permeability

The bidirectional MDR1-MDCK permeability was studied for compounds in the absence and presence of P-glycoprotein inhibitor, amprenavir. MDR1-MDCK monolayers were grown to confluence on collagen-coated, microporous, polycarbonate membranes in 12-well Costar Transwell plates. The assay buffer was Dulbecco's modified Eagle medium (DMEM). The dosing solution concentrations were 2 μM for the test compounds and 3 μM for amprenavir, prepared in DMEM. The receiver chambers were filled with blank DMEM. Cell monolayers were dosed on the apical side (A-to-B) or basolateral side (B-to-A) with test compound and incubated at 37°C with 5% CO_2 in a humidified incubator. Each determination was performed in duplicate. After 90 min, aliquots were taken from the receiver and donor chambers. These samples were mixed, diluted with one volume of water, and then assayed by LC–MS/MS using electrospray ionization. The lucifer yellow flux was measured for each monolayer after being subjected to the test compound to ensure no damage was inflicted to the cell monolayers during the flux period.

Rat Plasma Protein Binding

Protein binding was evaluated using rapid equilibrium dialysis. Plasma (Sprague–Dawley rat) spiked with test compound (1.0 $\mu\text{g}/\text{mL}$) was

added to the donor side of each dialysis chamber, and phosphate-buffered saline (PBS) was added to the receiver side of each dialysis chamber ($n = 2$). The dialysis chambers were sealed and placed in an enclosed heated rocker and equilibrated for 4 h at 37 °C. At 4 h, the matrix and PBS chambers were sampled and analyzed by LC/MS/MS.

Animal Studies

All studies were conducted according to GSK's Policy on the Care, Welfare and Treatment of Lab Animals and reviewed by the Institutional Animal Care and Use Committee at GSK, protocol IDs AUP0293 and AUP0448.

The in vivo pharmacokinetic profiles of compounds in Table 3 were determined in the Han Wistar rats (Charles River). The compounds ($n = 3$) were dosed as an IV infusion over 30 min, and blood samples were collected over a 24 h period and analyzed by LC/MS/MS to acquire concentrations. Subsequent analysis of blood concentrations with Phoenix WinNonlin software yielded the pharmacokinetic parameters presented in Table 3.

IV Dose Response and mRNA Expression of 5

Han Wistar rats (Charles River) were administered 5 by IV infusion over 6 h, and lung and kidney tissues were analyzed for expression of NRF2-regulated genes. Six animals per group received an IV infusion of compound 5 in vehicle (25 mM tris-buffered saline, pH 7.8) delivered at 1.5 mL/h at various concentrations to achieve a final, targeted total delivered dose of: 1, 10, 30, and 100 $\mu\text{mol/kg}$ in one experiment and at 5 $\mu\text{mol/kg}$ in a separate experiment. Two animals per group received vehicle only. Lungs and kidneys were collected following the infusion, and a portion of the tissue was stabilized in RNAlater for analysis of mRNA expression.

Rat Paraquat Model: Intratracheal Administration of 5

Micronized 5 was diluted in PBS with 0.5% Tween 80 and subsequently administered at 0.3, 1, 3, and 10 $\mu\text{mol/kg}$ in a dose volume of 67 $\mu\text{L/kg}$. Rats were anesthetized in a small acrylic induction box with 2–5% isoflurane gas. When surgical anesthesia was obtained (assessed by loss of the righting reflex followed by use of the pedal withdrawal reflex), the rat was placed supine on a head up, tilted platform. The trachea was illuminated, and dosing solution or vehicle was injected via a blunted 1.5-in. needle, just anterior to the carina. The rat was immediately returned to the recovery cage, where righting reflex was regained in 2–3 min. Twenty four hour later, paraquat (0.05 mg/kg, PQ or N,N' -dimethyl-4,4'-bipyridinium dihydrochloride, Sigma-Aldrich, St Louis, MO) was prepared in sterile PBS and instilled directly to the trachea of the rat while under isoflurane anesthesia. Rats were anesthetized in the same manner as above. The trachea was illuminated, and either PQ (0.05 mg/kg) or saline vehicle (300 μL) was directly instilled into the trachea, anterior to the primary bifurcation at the carina, using a blunt-tipped needle. The animal recovered as above. At 48 h postparaquat, rats were euthanized with 0.5 mL Fatal Plus IP and exsanguinated via cardiac puncture with a heparinized syringe. Lungs were removed en bloc and were cleaned of excessive tissue. The lung tissue was fast frozen and used for NQO1 gene expression and NQO1 activity and total glutathione analysis.

RNA Extraction and Real-Time PCR Analysis for In Vivo Studies

Total RNA was isolated using GE Healthcare RNeasy Mini RNA isolation kit (GE Healthcare, USA) or Trizol reagent (Life Technologies) according to the manufacturer's instructions and quantified using an Agilent RNA 6000 Nano instrument. Real-time PCR was performed on the ABI Prism 7900HT sequence detector (Applied Biosystems) using 18S rRNA as the internal control. The cDNA was amplified with specific primers for NQO1/Nqo1 (Rn00566528_m1), TXNRD1/Txnrd1 (Rn00581106_m1), SRXN1/Srxn1 (Rn01536084_g1), and 18S RNA (4310893E) using universal master mix (Applied Biosystems). The primers/probes sets used for amplification of cDNA were obtained from TaqMan Gene Expression Assays (Applied Biosystems). An identical threshold cycle (CT) was applied for each gene of interest. The comparative CT ($\Delta\Delta\text{CT}$) relative quantification method was used to calculate the

relative mRNA levels of target genes as described in the Applied Biosystems Chemistry Guide.

Glutathione Analysis

Total glutathione levels were measured from deproteinized samples using a commercially available assay (Cayman Chemical, Ann Arbor, MI) with absorbance detected at 405 nm (BioTek Powerwave HT plate reader/Gen 5.0 software, Winooski, VT).

NQO1 Analysis

NQO1-specific enzyme activity of rat lung tissue was measured using an assay adapted from Prochaska et al. that has previously been described.^{24,27}

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.6c01820>.

Protein–ligand crystallography data collection and refinement statistics; $2F_o - F_c$ map and omit map electron density for the KEAP1-5 crystal structure; SPR data for 5 binding to the Kelch domain of KEAP1; refinement summary and crystallographic data tables for the small-molecule crystal structure of 5; NMR data for 5 to support conformational analysis; proteins affected by 5 in affinity-enrichment-based chemo-proteomics (AEC) and TPP experiments; ^1H NMR binding data of 5 to KEAP1 and KLHL2 Kelch domains; LC–MS data for compound 5; VCD analysis; compounds 36-S, 27, and 5; and preparation of 7–21 and 23–24 (PDF)

Molecular formula strings and data (CSV)

Accession Codes

Deposition of Crystallographic Data: coordinates and structure factors for the KEAP1/Kelch-5 complex have been deposited with PDB with accession code: 3OHE. The crystallographic information file for the small molecule structure of 5 has been deposited with the Cambridge Crystallographic Data Centre (CCDC 2541002). This data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/structures. Authors will release the atomic coordinates and experimental data upon article publication.

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Notes

The authors declare the following competing financial interest(s): Torren Peakman is a consultant for Astex Pharmaceuticals; all other authors are current or former employees of GSK or Astex Pharmaceuticals, which have a commercial interest in the development of these compounds.

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ABBREVIATIONS

AEC	affinity enrichment chemo-proteomics
AKR1C3	aldo-keto reductase family 1, member C3
ALI	acute lung injury
AMPA	artificial membrane permeability assay
APRT	adenine phosphoribosyltransferase
ATCC	American type culture collection
BSEP	bile salt export pump
BTB	broad-complex, tram track and bric-a-brac
CETSA	cellular thermoshift assay
Chrom log D	chromatographic log D
Cl	clearance
COASY	CoA synthase
COPD	chronic obstructive pulmonary disease
CPOX	coproporphyrinogen oxidase
Cul3	cullen3
DCM	dichloromethane
DEA	diethylamine
DMEM	Dulbecco's modified Eagle medium
DMSO	dimethyl sulfoxide
dppf	1,1'-bis(diphenylphosphino)ferrocene
EDTA	ethylenediaminetetraacetic acid
Et ₃ N	triethyl amine
EtOAc or EA	ethyl acetate
FABP1	fatty acid-binding protein 1 liver
FBXL17	F-box and leucine-rich repeat protein 17
FP	fluorescence polarization
GSH	glutathione
HO-1	heme oxygenase 1
HPLC	high-performance liquid chromatography
HSQC	heteronuclear single quantum coherence

IPA	isopropanol
ITC	isothermal scanning calorimetry
IV	intravenous
IVR	intervening region
KEAP1	kelch-like ECH-associated protein 1
KLHL	kelch-like protein
KOtBu	potassium <i>tert</i> -butoxide
LC-MS	liquid chromatography–mass spectroscopy
MDAP	Mass-directed auto prep
MDCK	Madin–Darby canine kidney
MDR-1	multidrug resistance-1
MeOH	methanol
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
NQO1	NAD(P)H dehydrogenase [quinone] 1
nBu ₃ P	tri- <i>n</i> -butylphosphine
NRF2	nuclear factor erythroid 2-related factor 2
OATP1B1	organic anion transporting polypeptide 1B1
PCR	polymerase chain reaction
PDE	phosphodiesterase
PE	petroleum ether
PGAM5	phosphoglycerate mutase family member 5
PK	pharmacokinetic
PPB	plasma protein binding
PPOX	protoporphyrinogen oxidase
PQ	paraquat
PPI	protein–protein interaction
ROS	reactive oxygen species
ROESY	rotating-frame nuclear Overhauser effect spectroscopy
SFC	supercritical fluid chromatography
SQSTM1	sequestosome-1
SRXN1	sulfiredoxin-1
TFA	trifluoroacetic acid
THF	tetrahydrofuran
TLC	thin-layer chromatography
TPP	thermal proteome profiling
TR-FRET	time-resolved fluorescence energy transfer
TUBB	tubulin, beta class I
TXNRD1	thioredoxin reductase 1
UV	ultraviolet
VCD	vibrational circular dichroism
YIPF3	Yip1 family member 3
V _{ss}	volume of distribution at a steady state

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